

Confidence Updating for the PLN AND Rule: A Weakness-Composition Approach via Information Quantale

Abstract

This document presents a principled framework for updating confidence values in the PLN AND rule using weakness composition and information-theoretic quantales. We derive formulas for structural weakness (w_{struct}) and overlap factors from data, and establish the relationship between PLN and NARS approaches to conjunction.

1. Introduction

The PLN (Probabilistic Logic Networks) AND rule combines two statements A and B to derive their conjunction $A \wedge B$. The key challenge is determining the confidence of the conjunction given the confidences of the components. We present a principled approach based on:

1. **Weakness composition** from quantale theory
2. **Information-theoretic measures** for structural weakness and overlap
3. **Data-driven estimation** when empirical data is available
4. **Unified PLN-NARS perspective** showing both as computational strategies

2. Mathematical Framework

2.1 The Evidence-Weakness Quantale

Following the unified quantale framework, we work with:

$$Q_E = (E \times W, \leq, \otimes, I)$$

Where:

- **E** = $[0, \infty)$: Evidence/information space
- **W** = $[0, 1]$: Weakness (probability of error)
- **Confidence**: $c = 1 - w = e/(e+k)$ where k is a system parameter
- **Order**: $(e_1, w_1) \leq (e_2, w_2)$ iff $e_1 \leq e_2$ and $w_1 \geq w_2$

2.2 Information Quantale Transformation

Define the information quantale via logarithmic transformation:

$$Q_I = (I \times W, \leq_I, \otimes_I, 1_I)$$

Where:

- **I = -log(P)**: Information content (self-information)
- **Transformation**: $i(p) = -\log p$, $p(i) = e^{(-i)}$
- **Monoidal operation**: $(i_1, w_1) \otimes_I (i_2, w_2) = (i_1 + i_2, w_1 \oplus w_2)$

2.3 Weakness Composition

The fundamental weakness composition operation (corrected for probabilistic OR):

$$w_1 \oplus w_2 = 1 - (1-w_1)(1-w_2)$$

This represents the union of failure probabilities.

3. PLN AND Rule Confidence Updating

3.1 Problem Statement

Given:

- Statement A with strength s_A and confidence c_A
- Statement B with strength s_B and confidence c_B
- Context C (universe of discourse)

Derive: $A \wedge B$ with strength $s_{A \wedge B}$ and confidence $c_{A \wedge B}$

3.2 Strength Calculation

Under independence assumption:

$$s_{A \wedge B} = s_A \times s_B$$

Under dependence with correlation q :

$$s_{A \wedge B} = s_A \times s_B + q\sqrt{(s_A(1-s_A) \times s_B(1-s_B))}$$

3.3 Confidence via Weakness Composition

The confidence update follows three steps:

Step 1: Convert to weaknesses

$$w_A = 1 - c_A = k/(e_A + k)$$

$$w_B = 1 - c_B = k/(e_B + k)$$

Step 2: Compute structural weakness

$$w_{\text{struct}} = 1 - P(A \cap B | A \vee B) \quad // \text{ How well A,B jointly cover their union}$$

Step 3: Compose weaknesses

$$w_{A \wedge B} = 1 - (1 - w_A)(1 - w_B)(1 - w_{\text{struct}})$$

$$c_{A \wedge B} = 1 - w_{A \wedge B}$$

3.4 The Overlap Factor Refinement

To avoid double-counting evidence:

$$c_{A \wedge B} = c_A \times c_B \times \text{overlap_factor} / \text{normalization}$$

Where $\text{overlap_factor} \in [0,1]$ measures evidence independence.

4. Information-Theoretic Derivation

4.1 Structural Weakness via Entropy

Theorem 1 (Information-Theoretic Structural Weakness)

$$\begin{aligned} w_{\text{struct}}(A,B) &= H(A \wedge B | A \vee B) / H(A \wedge B) \\ &= 1 - I(A;B) / \min(H(A), H(B)) \end{aligned}$$

Proof sketch:

1. In information space, weakness corresponds to conditional entropy
2. $H(A \wedge B | A \vee B)$ measures uncertainty in conjunction given disjunction
3. Normalizing by $H(A \wedge B)$ gives relative weakness
4. This equals $1 - \text{normalized mutual information}$

4.2 Overlap Factor via Mutual Information

Theorem 2 (Information-Theoretic Overlap)

$$\text{overlap_factor} = \exp(-\lambda \times I(\text{evidence_A}; \text{evidence_B}) / H_{\text{min}})$$

Where:

- $\lambda \in [0,1]$ is a scaling parameter (typically 1)
- $H_{\min} = \min(H(\text{evidence_A}), H(\text{evidence_B}))$
- evidence_X refers to the observational support for X

Justification:

- Independent evidence: $I = 0 \rightarrow \text{overlap} = 1$ (full multiplication)
- Complete dependence: $I = H_{\min} \rightarrow \text{overlap} = e^{(-\lambda)}$ (maximal reduction)
- Exponential decay interpolates smoothly

4.3 Complete Information-Quantale Formula

Combining the above:

Algorithm: PLN-AND-Confidence-Update

Input: $(s_A, c_A), (s_B, c_B)$, context C, data D

1. Extract evidence counts:

$$e_A = c_A \times k / (1 - c_A)$$

$$e_B = c_B \times k / (1 - c_B)$$

2. Estimate from data (if available):

$$H_A = -\sum P(a) \log P(a) \text{ // Entropy of A's distribution}$$

$$H_B = -\sum P(b) \log P(b) \text{ // Entropy of B's distribution}$$

$$I(A;B) = \sum P(a,b) \log(P(a,b)/(P(a)P(b)))$$

3. Compute structural weakness:

$$w_struct = 1 - I(A;B) / \min(H_A, H_B)$$

4. Compute overlap factor:

$$\text{overlap} = \exp(-I(\text{evidence_A}; \text{evidence_B}) / \min(H(\text{evidence_A}), H(\text{evidence_B})))$$

5. Compose confidence:

$$w_combined = 1 - (1-w_A)(1-w_B)(1-w_struct)$$

$$c_A \wedge B = c_A \times c_B \times \text{overlap} / (1 + k \times w_combined)$$

6. Return $(s_A \wedge B = s_A \times s_B, c_A \wedge B)$

5. Estimation from Data

5.1 Empirical Entropy Estimation

Given samples $D = \{(a_i, b_i)\}_{i=1}^n$:

$$\hat{H}(A) = -\sum_a (n_a/n) \log(n_a/n)$$

$$\hat{H}(B) = -\sum_b (n_b/n) \log(n_b/n)$$

$$\hat{I}(A;B) = \sum_{\{a,b\}} (n_{\{ab\}}/n) \log(n \times n_{\{ab\}}/(n_a \times n_b))$$

With finite sample correction (Miller-Madow):

$$\hat{H}_{\text{corrected}}(A) = \hat{H}(A) + (|A|-1)/(2n)$$

5.2 Continuous Variable Estimation

For continuous variables, use:

Kernel Density Estimation:

$$\hat{p}(x) = (1/nh) \sum_i K((x-x_i)/h)$$

$$\hat{H} = -\int \hat{p}(x) \log \hat{p}(x) dx$$

k-Nearest Neighbor Estimation:

$$\hat{I}(A;B) = \psi(k) - \langle \psi(n_x + 1) + \psi(n_y + 1) \rangle + \psi(n)$$

Where ψ is the digamma function.

5.3 Evidence Overlap Estimation

To estimate overlap between evidence sets:

Algorithm: Estimate-Evidence-Overlap

Input: evidence_samples_A, evidence_samples_B

1. Compute support overlap:

$S_A = \{\text{indices where A was observed}\}$

$S_B = \{\text{indices where B was observed}\}$

$jaccard = |S_A \cap S_B| / |S_A \cup S_B|$

2. Compute statistical dependence:

$q = \text{distance_correlation}(\text{evidence_A}, \text{evidence_B})$

3. Combine measures:

$\text{overlap} = (1 - jaccard/2) \times (1 - q^2/2)$

Return: $\text{overlap} \in [0,1]$

6. Relationship to NARS

6.1 NARS AND Rule

In NARS, the conjunction $A \wedge B$ from statements $A (f_A, c_A)$ and $B (f_B, c_B)$ uses:

$f_{A \wedge B} = f_A \times f_B$

$c_{A \wedge B} = c_A \times c_B \times (f_A + f_B - f_A \times f_B)$

6.2 Unified Perspective

Key Insight: PLN and NARS differ primarily in their handling of context:

- **PLN:** Maintains explicit term probabilities $P(A|C)$, $P(B|C)$
- **NARS:** Uses direct projection without term probabilities

Under high uncertainty (uniform term probabilities):

$\text{PLN confidence} \approx \text{NARS confidence}$

6.3 Mapping Between Systems

The NARS formula can be derived from PLN by setting:

1. Uniform priors: $P(A|C) = P(B|C) = 0.5$
2. $w_{\text{struct}} = 1 - (f_A + f_B - f_A \times f_B)$ (NARS's overlap factor)
3. No additional evidence overlap correction

Derivation:

Starting from PLN:

$$c_{A \wedge B} = c_A \times c_B \times \text{overlap} / (1 + k \times w_{\text{struct}})$$

With NARS assumptions:

$$\text{overlap} = f_A + f_B - f_A \times f_B \quad (\text{inclusion-exclusion})$$

$$k \rightarrow 0 \quad (\text{high confidence regime})$$

Yields:

$$c_{A \wedge B} = c_A \times c_B \times (f_A + f_B - f_A \times f_B)$$

This matches the NARS formula exactly.

7. Practical Implementation Guidelines

7.1 When to Use Which Approach

Use NARS-style (simple product with overlap):

- Few samples ($n < 10$)
- No access to underlying data distribution
- Real-time inference requirements
- High uncertainty about term probabilities

Use PLN with full information-theoretic estimation:

- Sufficient data ($n > 100$)
- Access to joint distributions
- Batch processing acceptable
- Need for precise confidence calibration

Use hybrid (Jeffrey's rule combination):

- Moderate data ($10 \leq n \leq 100$)
- Mixed evidence quality
- Some distributional information available

7.2 Parameter Selection

System parameter k:

$$k = \sqrt{(\text{average_domain_cardinality})}$$

Typically $k \in [1, 10]$ for most domains.

Scaling parameter λ for overlap:

- $\lambda = 1$: Standard exponential decay
- $\lambda = 0.5$: Gentler penalty for correlation
- $\lambda = 2$: Aggressive independence requirement

7.3 Computational Considerations

Complexity:

- Entropy estimation: $O(n \log n)$ for sorting
- Mutual information: $O(n^2)$ naive, $O(n \log n)$ with KD-trees
- Overall AND update: $O(n \log n)$ with proper implementation

Numerical Stability:

- Use log-space computation for products
- Add small $\epsilon = 10^{-10}$ to avoid $\log(0)$
- Clip final confidence to $[0, 1-\epsilon]$

8. Example Calculation

Consider combining:

- A: "bird is red" with $s_A = 0.3$, $c_A = 0.8$
- B: "bird can fly" with $s_B = 0.9$, $c_B = 0.6$

Step 1: Evidence extraction ($k=1$)

$$e_A = 0.8 \times 1 / (1 - 0.8) = 4$$
$$e_B = 0.6 \times 1 / (1 - 0.6) = 1.5$$

Step 2: From data, suppose we find:

$$I(\text{red}; \text{flies}) / \min(H(\text{red}), H(\text{flies})) = 0.1 \text{ (weak correlation)}$$
$$\rightarrow w_{\text{struct}} = 1 - 0.1 = 0.9$$

Step 3: Evidence overlap:

Assuming independent observations:

$$\text{overlap} = \exp(-0) = 1.0$$

Step 4: Confidence composition:

$$w_A = 0.2, w_B = 0.4$$

$$w_combined = 1 - (1-0.2)(1-0.4)(1-0.9) = 1 - 0.8 \times 0.6 \times 0.1 = 0.952$$

$$c_A \wedge B = 1 - 0.952 = 0.048$$

With normalization:

$$c_A \wedge B = 0.8 \times 0.6 \times 1.0 / (1 + 1 \times 0.952) = 0.48 / 1.952 \approx 0.246$$

9. Conclusions

This framework provides:

1. **Principled foundation** via quantale theory and information geometry
2. **Data-driven estimation** of structural weakness and overlap
3. **Unified PLN-NARS view** as different points in the same framework
4. **Practical algorithms** for confidence updating

The key insight is that confidence updating for conjunction requires tracking both the structural relationship between concepts (via w_struct) and the independence of evidence (via overlap factor), both naturally expressed in the information quantale.

References

- Wang, P. (2006). *Rigid Flexibility: The Logic of Intelligence*
- Goertzel, B. et al. (2008). *Probabilistic Logic Networks*
- Cover, T. & Thomas, J. (2006). *Elements of Information Theory*
- Rosenthal, K. (1990). *Quantales and their Applications*